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Department of Earth Science and Engineering

MSc in Geo-Energy with Machine Learning and Data Science

Independent Research Project

Project Report

Developing Machine Learning Methods for Crystal Texture Analysis to Investigate Magmatic Evolution and the Onset of Explosive Volcanic Eruptions

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Abstract

Traditional petrographic analysis of plagioclase crystal textures which are indicators of processes linked to explosive volcanism (i.e., magma mixing, ascent and recharge) are vulnerable to human error, observer bias and are time consuming. Successful machine-learning-based automation of such analysis would significantly reduce bias and inaccuracy, increase the speed and scope of studies and offer insights into unseen textural relationships within crystal populations. This study applies self-supervised learning (SSL) to a segmented large unlabelled dataset of raw cold-cathodoluminescence (Cold-CL) scans. Cold-CL scans of entire thin-sections are segmented into individual plagioclase crystal images and are then combined into a unified dataset on which four suitable SSL baseline models (SimCLR, SimSiam, SwAV, DeepCluster) are trained for 100 epochs. Optimised configurations of the two best performing baselines (SimSiam and SwAV) were found by running 30 epoch grid searches with key hyperparameters. The configurations w05 and w12 (SwAV) and s10 (SimSiam) were identified as the best performing configurations outperforming equivalent baseline configurations. These three best configurations were then run for 100 epochs with w05 improving on the SwAV baseline with a peak median silhouette of 0.313 and w12 providing a strong SwAV alternative whereas s10 improved on some baseline metrics but remained relatively poor overall. When applied to unseen test data w05 performed to a similarly strong standard. Subsequent K-means clustering of SSL-learned embeddings produced texturally similar groups demonstrating that SSL could provide a powerful approach for automated analysis of crystal textures. With further development, SSL models applied to segmented crystal texture images may provide analysis comparable with a human observer at a fraction of the cost, time and at a much greater scale.

1. Introduction

Volcanic crystal textures preserve key information regarding magmatic processes and are used to understand magma storage conditions, mixing and ascent dynamics (Higgins, 2017; Moschini et al., 2023). Changes in crystal texture can be linked to processes triggering explosive eruptions (e.g. magma recharge, fluid exsolution). Plagioclase crystal morphology, zoning and population structure are linked to eruptive behaviour and so aid the hazard assessment of volcanoes (Streck, 2008; Haller, n.d.). Traditional petrographic analysis studies this key textural information using a human observer making this approach susceptible to human error, observer bias, variance between different observers and limited to small datasets due to lack of scalability. Therefore, this area of study would greatly benefit from the automation of such petrographic studies making them more cost-effective, scalable, efficient, accurate and unbiased. The use of machine-learning methods such as clustering would also make it possible to discover new relationships between textures and magmatic processes and further proof of existing ones.

Research utilising recent advances in the fields of machine learning, deep learning and data science have already begun to develop methods which push towards this aim. Some convolutional neural networks (CNNs) have been developed to segment crystals from thin section images such as cold-CL scans and produce data on properties such as shape, size and population distribution within images (Leichter et al., 2022; Hassan et al., 2024; Halverson et al., 2025; Morales et al., 2025). Adjacent studies have focused on producing fast, large-scale methods for otherwise human tasks such as analysing crystal populations, crystal size distribution and reducing the need for manual/human measurement and tracing (Jutzeler et al., 2024; Toth and Maclennan, 2024). These studies

demonstrate the somewhat latent capability of machine learning to greatly improve the scale, speed and accuracy of petrographic studies. Interestingly, most related studies focus on segmentation of crystal images, measurement and supervised classification. Few studies have worked on learning textural relations from large unlabelled crystal image collections. Self-supervised learning (SSL) can provide a solution to learning crystal textures from unlabelled cold-CL image data and to the lack of studies attempting to learn important textural data. SSL bypasses the lack of texture labels by creating a training signal from either transformed image views or automatically generated assignments instead of defined classes. Texturally related groups can then be identified from the subsequent clustering of learnt SSL outputs (embeddings). However, there can also be some key risks with this approach such as differing image geometry; data leakage between physically related samples; and shortcut learning issues derived from backgrounds, cropping and/or location-based characteristics instead of targeted crystal textures.

Although SSL has rarely been applied in petrology, it has been applied in other fields such as cell microscopy and histopathology where SSL has been shown to recover biologically meaningful structures without extensive human annotation (Ciga et al., 2020; Galusser et al., 2023; Bendidi et al., 2024; Moutakanni et al., 2025). Additionally, studies in more closely related fields like mining geology and palaeontology suggest learned visual representations are transferrable to other geoscience-related image analysis tasks (Ferreira, 2023; Koeshidayatullah and Ferreira, 2023). These studies suggest SSL methods do have significant potential in reading crystal textures in volcanic rocks.

This study aims to compare different SSL models for the clustering of volcanic, unlabelled cold-CL-sourced plagioclase images. This study has produced

PlagNetPlus, one of the first frameworks for the learning of such crystal-texture representations. PlagNetPlus attempts to mitigate some of the issues with SSL outlined above by producing a multi-location, canonical dataset, sample aware splitting, a common ResNet-18 backbone for all models and a common evaluation framework in which the test images are held-out throughout the study's development. 123,264 crystal images were segmented, assessed for quality and unified to produce a dataset spanning 27 samples from 5 distinct volcanic areas across the world (Aegina and Methana in Greece, Fish Canyon Tuff and Mt St Helens in the USA, Caraci in Romania). SSL methods were successfully tested and compared on a large crystal image dataset to determine the best-performing model for the outlined task. This work also suggests which SSL models are best for this task and points towards a direction for continued improvement in SSL model results for this and other closely related tasks.

2. Methodology

2.1 Initial Data and Segmentation Using PlagNet Package

Location	Samples	Number of Raw Scans	Proportion Raw Scans (%)	Notes
Aegina, Greece	**8 samples:** Aeg6, Aeg12, Aeg14, Aeg20, Aeg27, Aeg28, Aeg29, Aeg30	Already segmented	Already Segmented	-
Methana, Greece	**9 samples:** M83-27_IX, M83-7-1-(II), M83-7-1-II, M83-7-1-II_no_overlap, M91-3-10-, M91-8791-71-1, Met91-100-3(III), Met91-77-8-XI, Met91-78-9-XII	1,119	40.18%	M83-7-1-II_no_overlap identified as an altered duplicate of M83-7-1-II
Caraci, Romania	**3 samples:** 1b, 5, 10	489	17.56%	Named 'Rumania' in source folder.
Mt St Helens, USA	**2 samples:** 18_1a, 71512	312	11.20%	-
Fish Canyon Tuff (FCT), USA	**7 samples:** 6861_FCT, 6861_recoated, 7191, 7193, 83b, BIG', FNGR-2	865	31.06%	6861_recoated identified as an altered duplicate of 6861_FCT
Total	**29 samples:** (21 non-Aegina)	2,785	100%	These totals exclude Aegina.

Table 1 *– Breakdown of the source data for this study. Aegina data existed as pre-segmented plagioclase images. Sample names of duplicate samples are indicated in red. These were removed before segmentation.*

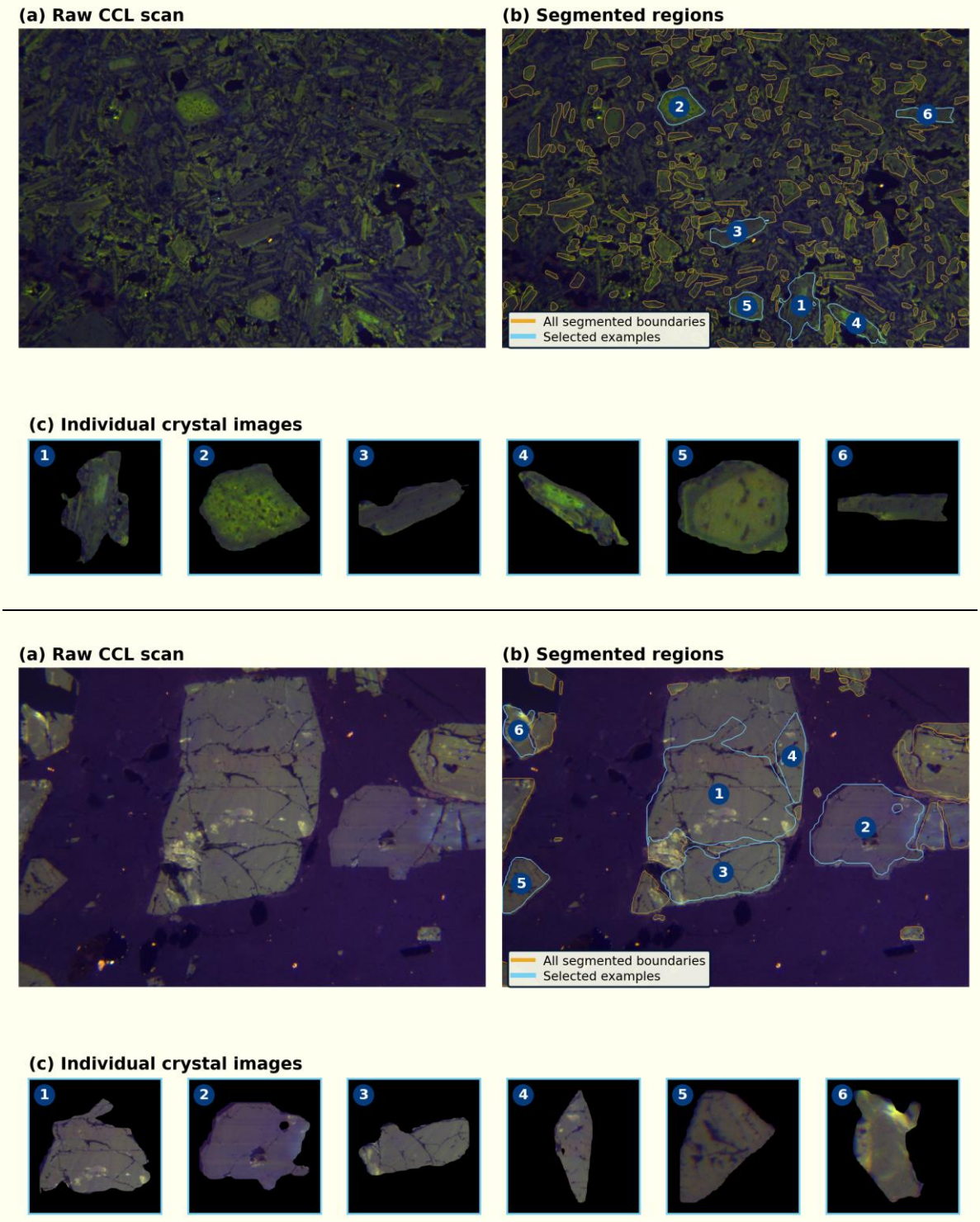


Figure 1 *- Examples of PlagNet-based segmentation applied to cold-CL scans: Both images highlight issues faced with accurate and complete segmentation. The top image (Methana - Met91-78-9-XII_00001_00002) shows how more complex groundmass causes crystals to remain unidentified and the bottom image (6861_00001_00002 - Fish Canyon tuff) shows how fracturing causes multiple segmentations in one crystal.*

Segmentation and production of the non-Aegina plagioclase images was completed using the PlagNet package (Nathwani and Peng, 2026) and crystal masks were used to generate the final images. Due to neighbouring tiles overlapping some crystals appeared more than once, so duplicate control rules were implemented allowing only one copy to be retained. Geometry checks highlighted 64 images with impossible (>1) crystal occupancy values (proportion of crystal inhabiting whole image). This issue was traced back to an inconsistency in how mask area and bounding box size had been calculated and was corrected. After these issues were fixed the maximum crystal occupancy across the dataset was ~0.971. Images that possessed potential issues were then reviewed visually by creating image contact sheets showing examples for random selections, low-confidence images (i.e., low segmentation score), images where crystals touch image borders, extremely small images and extremely elongated images. Some of the possibly anomalous images were reviewed and assigned with a 'PASS_WITH_NOTES' assessment which means crystals of uncertain quality are not completely removed but can be tracked if it is needed later to understand any issues.

2.2 Preprocessing after Segmentation

After the segmentation model completed segmenting all the cold-CL scans the segmentation masks were applied to the original cold-CL scans. From this RGB PNG plagioclase crystal images were created, padded into square images and given black backgrounds. Crystals were cropped by a 10% margin, subject to a minimum margin of 8 pixels to reduce edge noise. The padding and background colour preprocessing were then repeated for the already segmented Aegina crystal images.

2.3 Unified Dataset of Segmented Plagioclase Crystals

A segmented database inherited from PlagNet (Nathwani and Peng, 2026) of 12,344 pre-segmented plagioclase images from all eight Aegina samples was combined with the combined scans of 19 samples from Methana, Caraci, Mt St Helens and Fish Canyon Tuff. These were segmented into 110,920 images to create a unified dataset (plagnetplus_ssl_unified_v2) of 123,264 segmented plagioclase images from a combined 27 different samples across all five locations (Figure 2). In terms of proportion Methana provided the most plagioclase images with 68,008 followed by 21,305 from FCT, 18,506 from Romania, 12,344 from Aegina and 3,101 from Mt St Helens. 'M83-7-1-II_no_overlap' and '6861_recoated' were discovered as duplicate samples and removed before segmentation.

After the unified dataset plagnetplus_ssl_unified_v2 was created, a frozen benchmark called 'pooled_broad_p80_10_10_v2' containing 91,357 training crystals, 14,919 validation crystals and 16,988 test crystals was produced. This is a split of 74.11 : 12.1 : 13.78 (train : validation : test). It is split in this way because the target split was 80:10:10 but each crystal has a 'split_group_id' which allows crystals of the same sample to remain within either training, validation or test sets together with all other crystals of that sample. In this study a validation set refers to the set used for model evaluation during development whereas the test set was kept completely independent, reserved until a point where all models choices are decided and a final evaluation on unseen data is required. This is because crystals from the same physical samples are not truly independent and not doing this could lead to a model learning sample specific patterns. The 80:10:10 target split was chosen to give models as much training data as possible whilst retaining test and validation sets of a sufficient size. Natural frequency sampling (i.e., sampling according to image frequency within the original

dataset) was used for the baselines so each model has the same empirical distribution instead of model specific balancing schemes. Methana makes up 58.7% of the training split and therefore location probes are assessed using macro-F1 and a majority class baseline. If raw accuracy was used it would appear high because a classifier would repeatedly predict the dominant location. Mt St Helens only has two samples 71512 and 181a. All 3,017 crystals from 71512 contribute to the training set and all 84 images from 18_1a contribute to the test set. Therefore, before final testing all interpretations made are limited to sample 71512 and cannot assess generalisation to 18_1a.

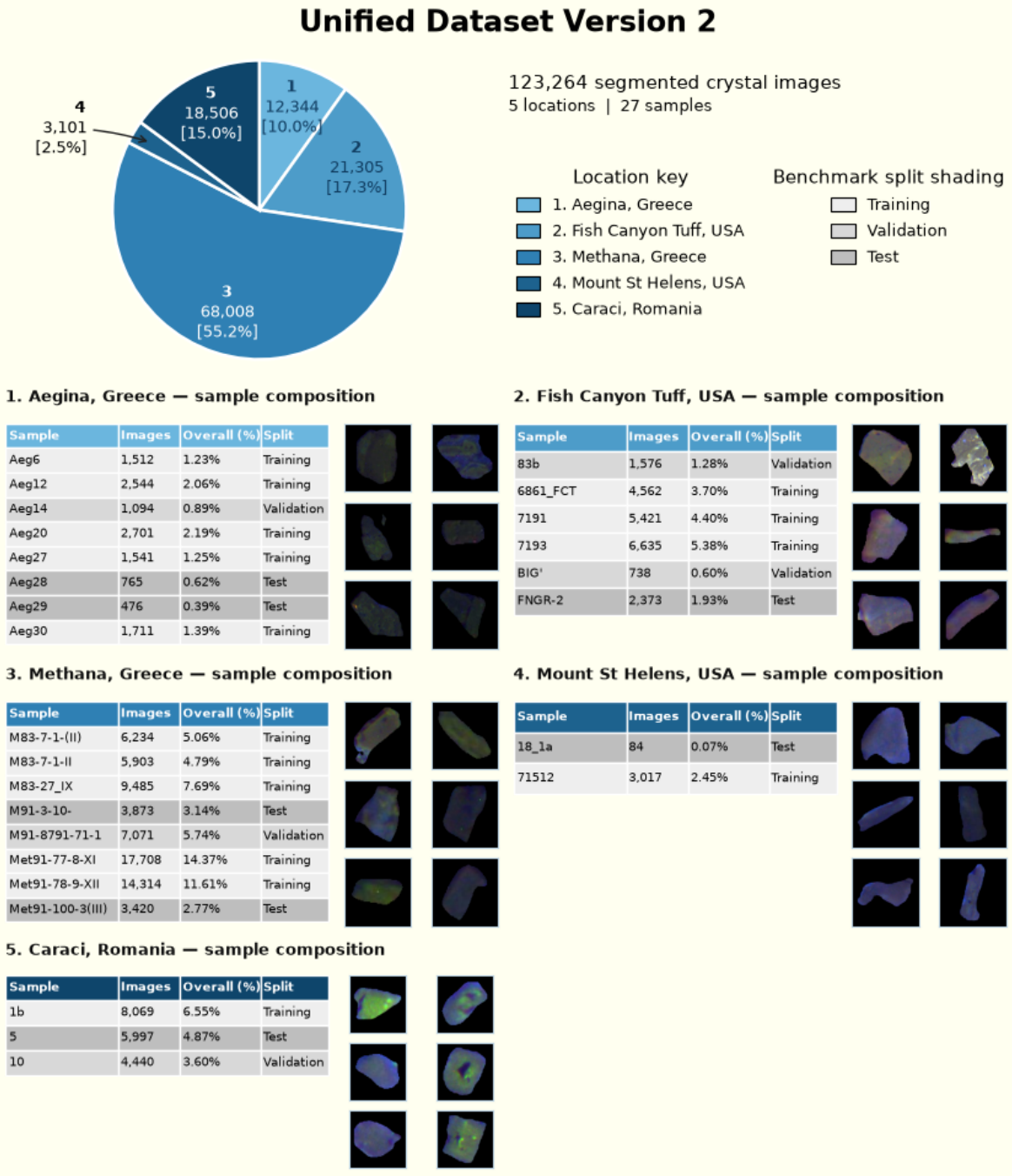


Figure 2 *– Summary of Unified Dataset V2. Pie chart shows proportion of dataset by location, lists showing samples by location and each samples image count and proportion of images within dataset. Shading shows which samples were used in which of the three sets (training, validation and test). Random examples of segmented images from each location are displayed alongside the table for the corresponding location.*

2.4 Baseline Models

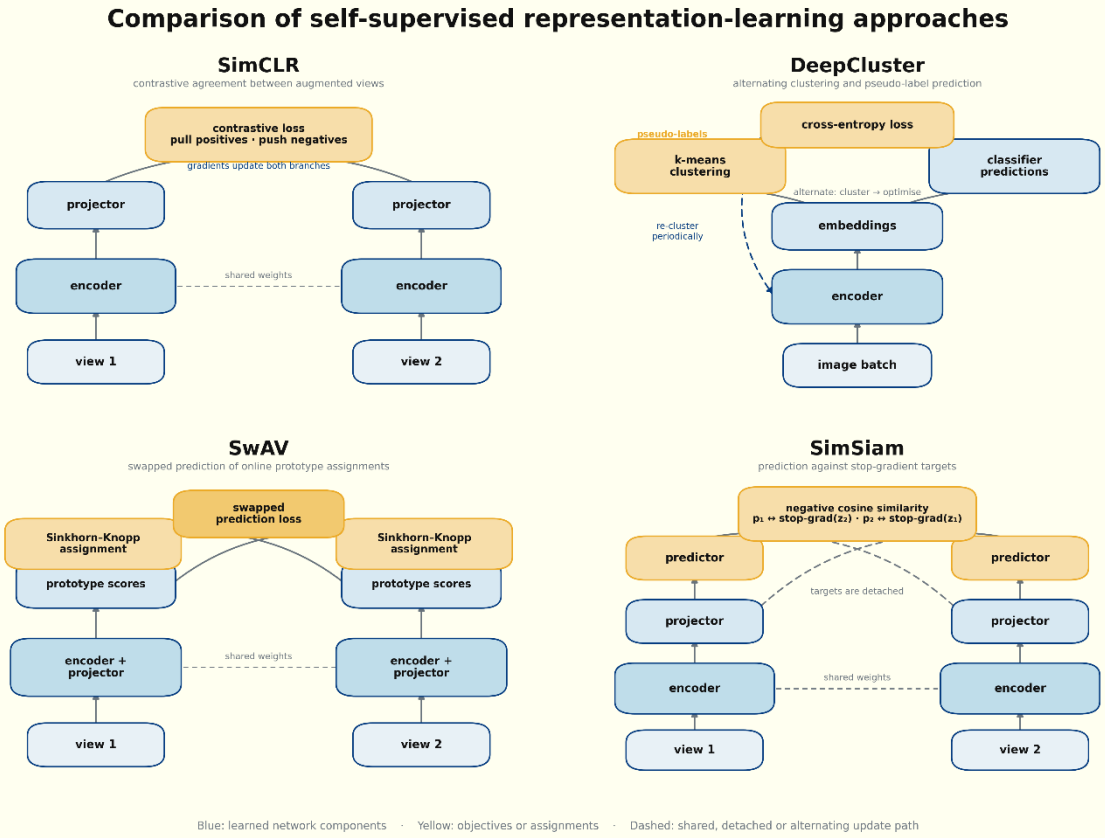


Figure 3 *– Model architecture comparison between the four chosen self-supervised representation learning models. SimCLR (simple framework for contrastive learning of visual representations) a contrastive learning model (top left), DeepCluster (deep clustering for unsupervised learning of visual features) a clustering-based model (top right), SwAV (swapping assignments between views) an online clustering-based model (bottom right) and SimSiam (simple Siamese representation learning) a non-contrastive Siamese model. Author's own figure; visual design adapted from Lee (2023).*

Four self-supervised representation learning models were selected as baselines. The settings for the baseline models provide fixed starting points and are not optimal but in turn the best performing models will be optimised using a grid search. Each is built on a common, randomly initialised PyTorch, torchvision ResNet-18 convolutional backbone that uses natural-frequency sampling, a

batch size of 256, eight data workers, random seed of 42 and gradient clipping at 5.0. However, they all differ in terms of how they generate a learning signal.

SimCLR is a contrastive learning method which learns by drawing augmented views of the same crystal closer together whilst separating representations of different crystals (Chen et al., 2020). The baseline SimCLR uses a 512 → 512 → 128 projection head and NT-Xent contrastive loss at a temperature of 0.2, trained for 100 epochs with AdamW (learning rate = 3e-4, weight decay = 1e-4). This creates a clear negative contrastive baseline (Chen et al., 2020).

DeepCluster is a clustering-based learning method that groups image features with k-means clustering and uses the resultant cluster assignments as temporary pseudo-labels for further training (Caron et al., 2018). Baseline DeepCluster alternates between deterministic feature extraction with PCA whitening to 128 dimensions, k-means clustering into 500 clusters, and one epoch of training with pseudo-labels, all repeated for 50 cycles (pseudo-labels are regenerated by re-clustering updated embeddings before every epoch training cycle). Inverse-cluster-frequency samples diminish any domination by larger provisional clusters. The classifier is reset after each re-clustering as the cluster indices do not keep a persistent meaning across cycles (Caron et al., 2018).

SwAV is a cluster-based learning method which learns assignments to a set of prototypes during training and encourages differing augmented views of the same crystal to predict one another's prototype assignments (Caron et al., 2020). Baseline SwAV uses a 2048 → 128 projection head, 500 learned prototypes, temperature = 0.1 and three Sinkhorn-Knopp iterations with epsilon set at 0.05. Two global views are implemented without a queue or local cropping which preserves the common two-view design. Risk of collapse (i.e. loss of representation diversity, where different inputs map to overly similar

embeddings) is lowered by having balanced prototype assignments, whilst each view can predict an assignment created by the other (Caron et al., 2020)

SimSiam is a non-contrastive Siamese method that learns agreement between two augmented views without needing negative examples like a contrastive method (Chen and He, 2021). Baseline SimSiam uses a three-layer 2048-dimensional projector and 2048 → 512 → 2048 predictor. Symmetric negative cosine similarity is calculated between each prediction and the stop-gradient representation (a projected representation used as a temporary target) of the opposite view. SGD is used with learning rate = 0.05, predictor learning rate = 0.05 and weight decay = 1e-4 for 100 epochs.

2.5 Model Input Preprocessing and Training

Images from the unified dataset were used to create two independently augmented views of each crystal used for most SSL methods implemented (independently augmented views are important for all models except DeepCluster). Before training every image is bicubically resized to 224 x 224. Image augmentations used include vertical and horizontal flipping (p = 0.5), random 0°, 90°, 180° or 270° rotations, brightness, contrast jitter (p = 0.8; strength = 0.2), Gaussian blur (p = 0.2; sigma = 0.1 - 2). Altering the orientation reduces the dependence on arbitrary crop alignment, while jitter at a moderate level reduces a model reliance on sample brightness which can vary between different acquisition settings. It was decided that saturation or hue modification would not be implemented to reduce the any risk of introducing colour signals that could disrupt the intended cold-CL textural information.

Mean and standard deviation values are calculated for each RGB colour channel using only training images these values are then used to normalise images before training. During training the images are randomly augmented but embeddings are extracted using fixed preprocessing each time to resize to 224 x 224 followed by conversion to tensors and normalised using training statistics. This creates reproducible embeddings and keeps them aligned with the dataset manifest (i.e., first embedding corresponds to first crystal in the manifest, the second embedding to the second crystal and so on).

Training was run using PBS jobs on Imperial College's CX3 HPC (high-performance computing) cluster. Each run used one NVIDIA L40S GPU, eight CPU cores and 64GB RAM. SwAV, SimSiam and SimCLR were all trained for 100 epochs with validation occurring after each epoch and checkpointing every ten epochs. DeepCluster ran for an equivalent 50 cycles. Each run configuration was recorded, and then 512-dimensional training and validation embeddings were exported back from the remote CX3 cluster.

2.6 Validation and Clustering

Each baseline model uses different training objectives so their training losses cannot be compared directly. Instead, they are compared by utilising the 512-dimensional embeddings produced by their trained ResNet-18 backbones. The embeddings are L2-normalised first so variations in feature magnitude do not dominate comparisons. They are then reduced to 50 principal components before clustering. Fifty was chosen in advance as a practical dimensionality reduction that would substantially lower computational cost whilst maintaining a relatively high-dimensional representation of the embedding space.

K-means clustering then divides validation embeddings into groups of similar crystals. The number of clusters, k, is tested for every integer value from 2 through to 200. Each value is run using three random seeds, 42, 43 and 44 to judge the reproducibility of the resulting clusters. Each fit uses all 14,919 validation embeddings. The Silhouette score which measures how clearly separated clusters are from one another (Table 2) is calculated on the same fixed subset of 5,000 validation crystals for every model and for each value of k. This validation subset is around a third of the total validation set and is used to reduce the computational cost of calculating Silhouette score whilst still being a large enough set to produce meaningful results. Crystals are kept the same in the subset to ensure results are directly comparable.

The preferred value for k is chosen by calculating the median silhouette score over the three random seeds and clustering stability measured with ARI (Adjusted Rand Index) to resolve any ties that might occur. The dense k search is more exploratory than a statistical test. This means the chosen k value is seen as the best performing cluster count using this validation procedure instead of being a definitive number of crystal populations. Silhouette and stability are considered along with the whole range of k values. Hyperparameters are chosen solely with the validation results. The test set is locked until a model setup is finalised after which it is run on the test set.

	What it Measures	Equations	Terms
Silhouette Score ↑	How well does each crystal fit its cluster relative to the next nearest cluster?	$s(i) = \frac{b(i) - a(i)}{max(a(i), b(i))}$	$a(i)$: mean distance within the assigned cluster, $b(i)$: mean distance to nearest alternative cluster.
Adjusted Rand Index (ARI) ↑	How consistent are cluster assignments between repeated k-means runs?	$ARI = \frac{\left\{X - \frac{AB}{T}\right\}}{\left\{0.5(A+B) - \frac{AB}{T}\right\}}$	X: pairs grouped together in both runs; A, B: paired assignments in each run; T: all possible pairs.
FCT cross-sample precision @10 ↑	How well are FCT samples retrieved from different physical samples? (FCT is an arbitrary location choice testing whether a model learns features shared across different samples from the same area)	$P@10_{FCT} = \frac{1}{\lvert Q_{FCT}\rvert} \sum_{q \in FCT} \frac{m_q}{10}$	Q_{FCT} : FCT query crystal set, q: one query crystal, m_q: number of its 10 nearest cross-sample neighbours from FCT. Crystals from query's own sample excluded.
Location-probe Macro F1 ↑	How effectively do the embeddings distinguish crystals from different geographic locations?	$Macro\ F1 = \frac{1}{C} \sum_{c=1}^{C} \frac{2P_c R_c}{P_c + R_c}$	C: number of locations (5 for this study); R_c and P_c : recall and precision for location c.
Source-Image Enrichment @10 ↓	Do nearest neighbours disproportionately originate from the same cold-CL scan?	$E@10 = \frac{p_{obs}}{p_{exp}}$	p_{obs} : observed retrieval of same scans; P_{exp} : expected retrieval by chance.
Effective Rank ↑	How many embedding dimensions contain variance that is meaningful?	$r_{\text{eff}} = \exp\left(-\sum_j p_j \ln p_j\right)$ where: $p_j = \frac{\lambda_j}{\sum_m \lambda_m}$	λ_j: variance along dimension j; P_j: proportion of total embedding variance; m: index across embeddings dimensions; λ_m: variance represented by dimensions m.

Table 2 *– metrics used for comparing model performance. Arrows convey whether lower or higher values indicate metric improvement.*

2.7 Grid Search Optimisation of Best Performing Baselines

After baseline model comparison only the two best models (SwAV and SimSiam) were selected for a grid search. The choice of using these two models is based on strong, complementary baseline results using somewhat different learning mechanisms and the need to cut the heavy grid search compute cost by half. The grid search explored the effects of changes to key SwAV and SimSiam hyperparameters.

For the SimSiam grid search 12 model configurations were tested by combining differing learning rates of 0.025, 0.05 and 0.1 with predictor learning rates of 0.025 and 0.05 and weight decay values of 1e-04 and 1e-5. A baseline equivalent configuration with a learning rate of 0.05, predictor learning rate of 0.05 and weight decay of 1e-04 was included within the grid so results can be compared to the baseline. Configurations trained for 30 epochs using the same random seed, batch size, augmentations and training split.

The SwAV grid search also had 12 configurations which encompassed changes in learning rates of 0.15, 0.3 and 0.6 mixed with prototype counts of 250 or 500 and temperatures of 0.05 or 0.1. all other settings were kept the same as baseline models. One of the SwAV configurations was also an equivalent to its baseline with a learning rate of 0.3, a prototype count of 500 and a temperature of 0.1. These configurations were also trained for 30 epochs each with the same random seed, batch size, augmentations and training split.

Only validation and training data was used during the grid search, and no test data was accessed. Each model's native loss was interpreted only for each model as the models optimise to different objectives. Therefore, SwAV was assessed using swapped prediction loss along with prototype concentration, prototype

occupancy and assignment behaviour. SimSiam was assessed using negative cosine similarity loss with collapse diagnostics.

2.8 Optimised Models Informed by Grid Search Results

The best configuration could not be selected on training loss alone as this is non-comparable between models. Furthermore, for SwAV this would not be appropriate as temperature and prototype count changes the scale and behaviour of the SwAV objective/loss (in all models used in this study the loss is mathematically the same as the objective). For SimSiam a decrease in validation loss (negative cosine similarity objective) is a sign of the two augmented views agreeing. This was considered together with collapse measurements and standard deviation of projection, so a configuration was not selected on loss alone. The configurations chosen (s10, w05 and w12) will then each be run for 100 epochs. The final optimised model will be tested using the unseen test data.

3. Results

3.1 Baseline Model Performance

Key Comparison Metrics	SimCLR	SimSiam	DeepCluster	SwAV
Median Silhouette Score ↑	0.0695	0.1479	0.1686	0.2939
Stability ARI (Adjusted Rand Index) ↑	0.6111	0.4750	0.6934	0.6230
FCT Cross-Sample P@10 ↑	0.0912	0.0884	0.1096	0.2362
Location-probe macro-F1 ↑	0.4718	0.6018	0.5395	0.4268
Source-Image Enrichment@10 ↓	2.1985	3.6455	4.1829	2.7456
Effective Rank ↑	33.6802	7.8902	29.3487	48.7477
GPU Hours ↓	8.0493	5.8022	3.2769	5.8433

Table 3 *– Baseline metrics across the four chosen models. Arrows indicate if a higher or lower score represents improvement for each metric. Model columns highlighted in yellow are those which were chosen as the best two baseline models)*

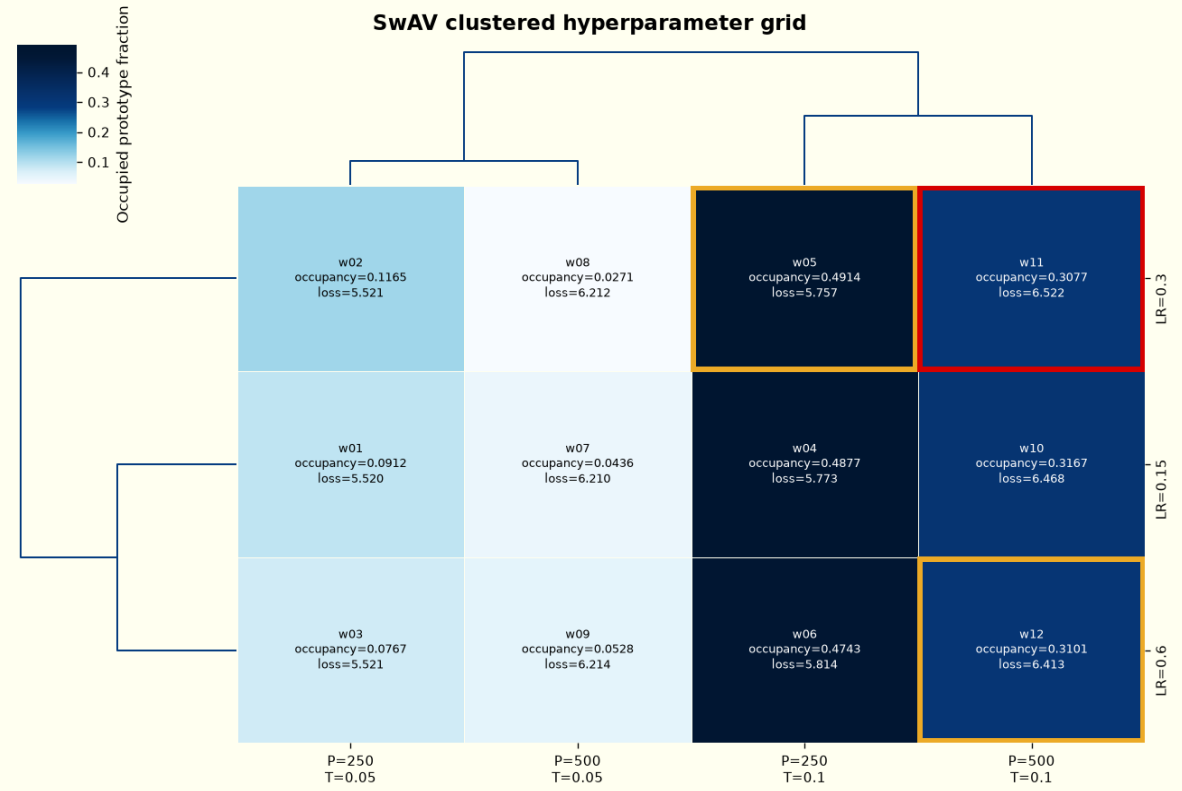
All four models produced quite different results even though they share a ResNet-18 backbone. The silhouette score was the highest for SwAV (0.294) which highlights SwAV as having the most defined embedding clustering. SwAV also had the highest FCT cross-sample precision@10 suggesting it has learned characteristics shared by FCT samples the most effectively. Effective rank also indicated SwAV was using a wider range of the 512-dimensional representation space compared to other baselines (SwAV effective rank = 48.8). Alternatively, SimSiam had an extremely low effective rank at 7.9 flagging it as collapse-prone and comparatively more compressed.

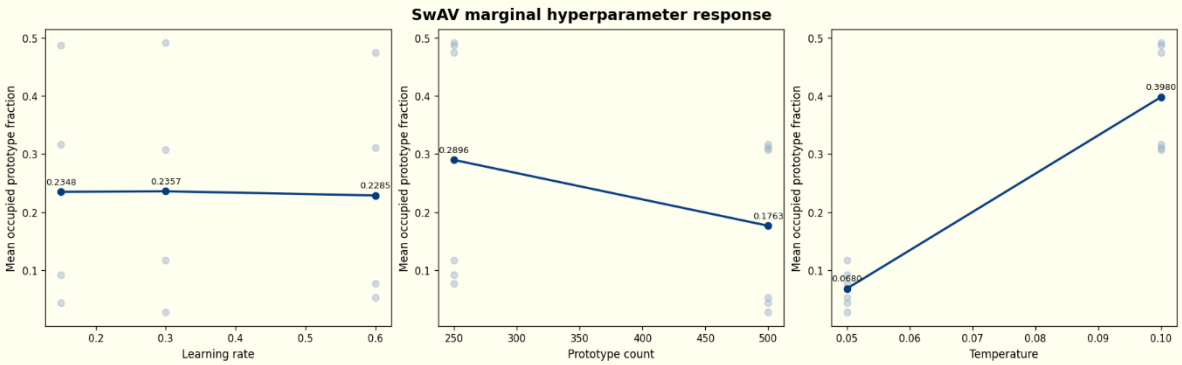
In terms of stability of clusters DeepCluster narrowly outperformed SwAV with an ARI score of 0.693 (SwAV ARI = 0.623). The ability of DeepCluster to produce

stable clusters without having clearly separated clusters (i.e., DeepCluster has a lower Silhouette than SwAV at 0.1686) highlights the importance of using multiple comparative metrics and shows SwAV's overall performance to be ahead of the other baselines.

SimSiam has significantly higher location probe macro-F1 (0.602) than other models. However, location score can be misleading. High scores could suggest the embeddings are easily distinguishing crystal textures from different locations but could also be falsely enhanced by other differences between locations such as learning variations in location scan brightness. As a result, this metric is diagnostic rather than a direct measurement of learning.

3.2 Insights from Grid Search Optimisation





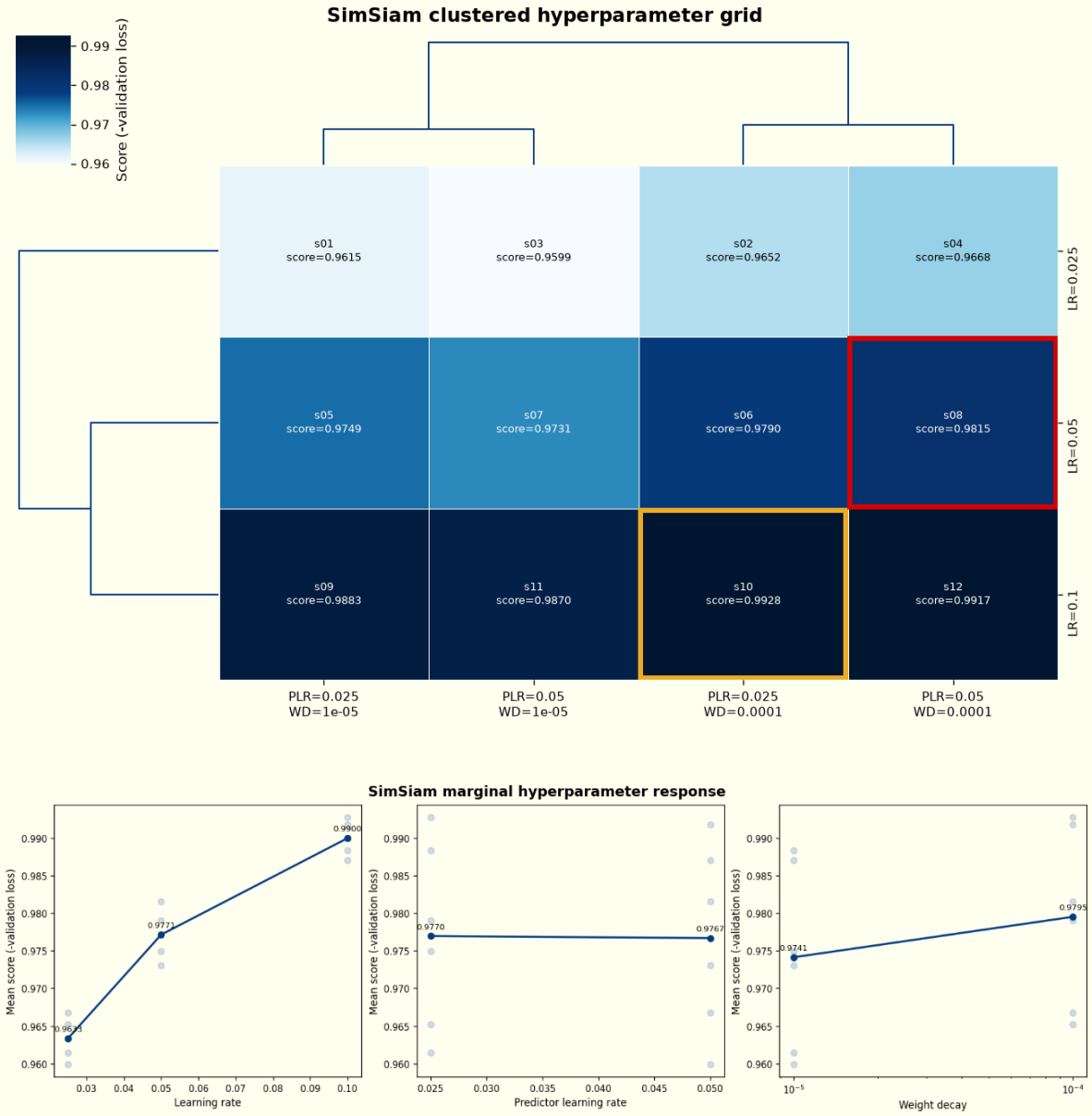


Figure 4 *– Grid search results for SwAV and SimSiam. The w05 and w12 configurations for SwAV and s10 for SimSiam which were chosen to run for 100 epochs due to good performance are highlighted in yellow. The baseline equivalent configurations w11 and s08 are highlighted in red. LR = learning rate, P = prototypes, T = Temperature, PLR = predictor learning rate, WD = Weight Decay. Author's own figure; plotting approach adapted from Lungu (2022).*

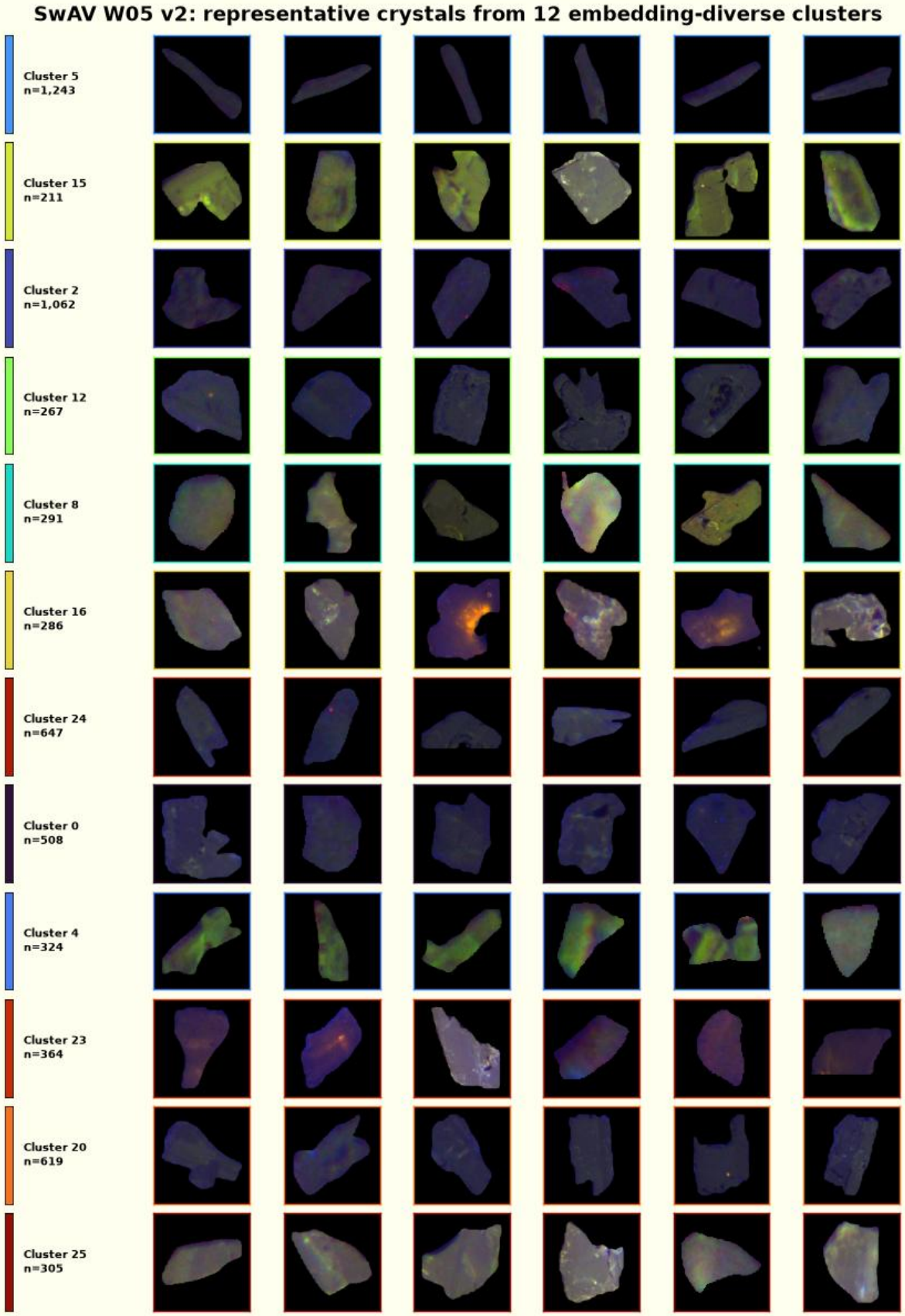
All 24 of the 30 epoch hyperparameter search runs were successful. When assessing the results prototype usage and loss had to be considered for SwAV. Configurations with a temperature value of 0.05 created lower loss values but

lower representation diversity was highlighted by assignments being concentrated into a small number of prototypes. When temperature was changed to 0.1 there was much wider, balanced prototype usage. The configuration w05 had the lowest validation loss (5.757) within the prototypes = 250 branch. Within the prototype = 500 branch w12 performed the best (loss = 6.413). Therefore, w05 and w12 were chosen as the best configurations from the SwAV grid search and were subsequently run for 100 epochs to produce accurate comparisons to the baselines. SimSiam configurations got better generally with increasing learning rate. All learning rate = 0.1 configurations outperformed any other configuration (in terms of loss) including the s08 baseline equivalent. The best overall SimSiam configuration was s10 (loss = -0.9928). Increases in learning rate seem more effective at improving results than increases in the other two parameters tested.

3.3 Optimised Model Performance and Testing Best Model

Key Comparison Metrics	SimSiam: s10	SwAV: w05	SwAV: w12
Median Silhouette Score ↑	0.1855	0.3139	0.3095
Stability ARI (Adjusted Rand Index) ↑	0.5678	0.8672	0.6990
FCT Cross-Sample P@10 ↑	0.1218	0.2209	0.2852
Location-probe macro-F1 ↑	0.5980	0.4616	0.4526
Source-Image Enrichment@10 ↓	2.5632	2.6022	2.5341
Effective Rank ↑	5.5739	35.4945	42.9081
GPU Hours ↓	7.543	7.418	7.443

Table 4 *- Optimised model metrics across the three chosen configurations (Best overall model highlighted in yellow). Arrows signify whether higher or lower values indicate metric improvement.*


SwAV W05 v2: representative crystals from 12 embedding-diverse clusters
Cluster 5
n=1,243
Cluster 15
n=211
Cluster 2
n=1,062
Cluster 12
n=267
Cluster 8
n=291
Cluster 16
n=286
Cluster 24
n=647
Cluster 0
n=508
Cluster 4
n=324
Cluster 23
n=364
Cluster 20
n=619
Cluster 25
n=305

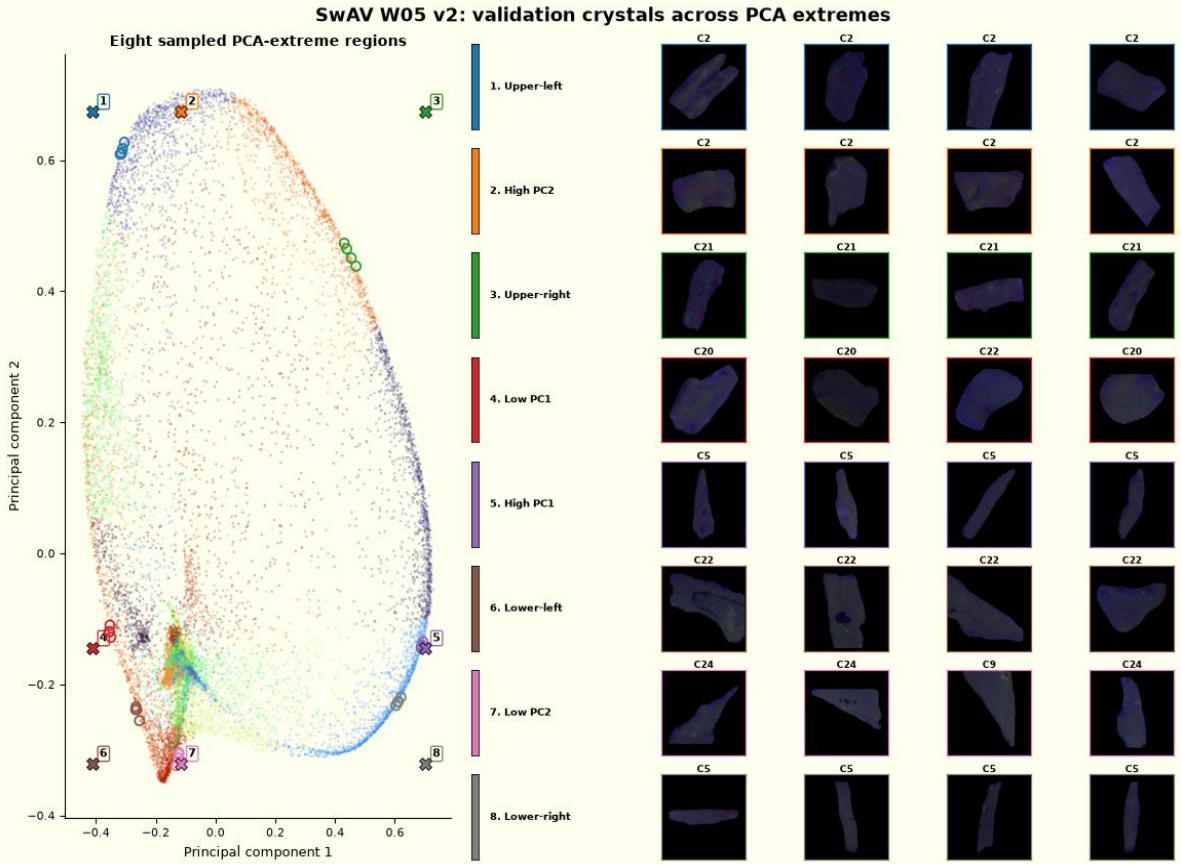
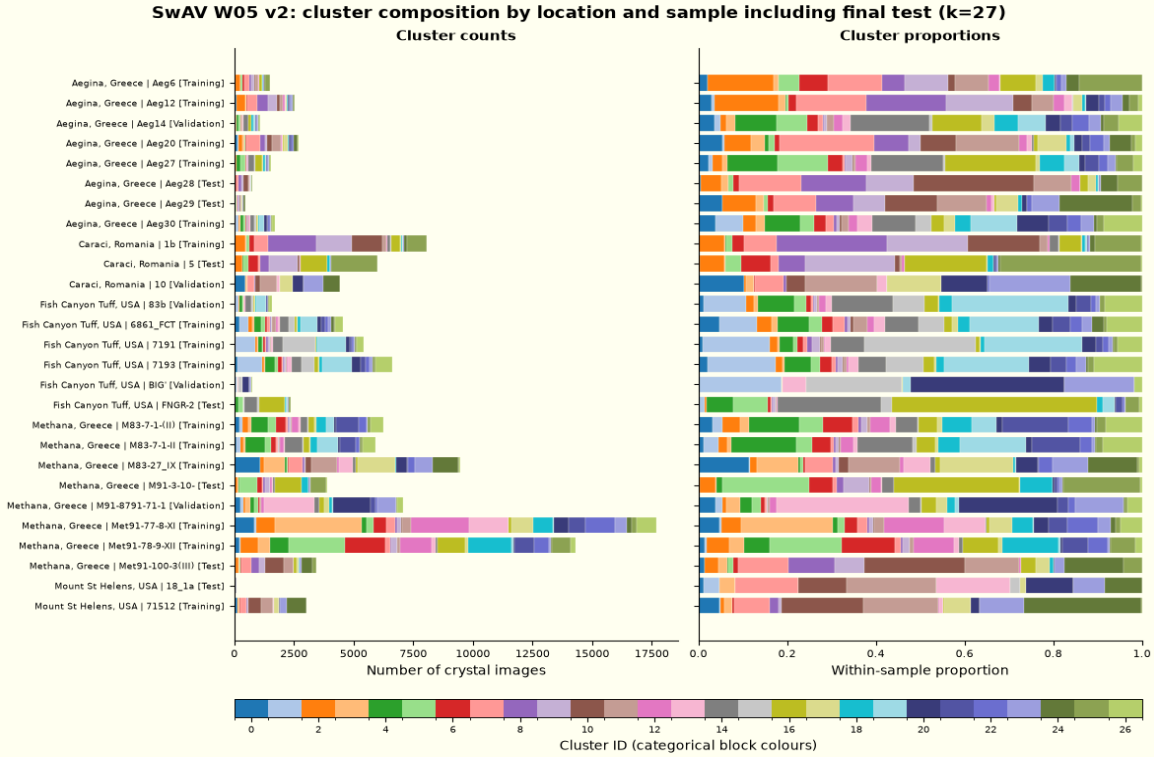


Figure 5 *– Clustering of learnt embeddings from the best performing model (SwAV: w05). Above is a representative contact sheet from 12 random clusters. Below is a contact sheet showing examples of crystals in the PCA extremes. Coloured crosses mark extremes and circles of the same colour show the nearest crystals to those extremes which are displayed (Based on w05 100 epoch run). The PCA extreme crystals are noticeably all dull/dark images. It seems there may be a loose trend to brightness/colour towards the centre of the plot.*

After the best configurations were compared the final model was chosen as w05 due to its strong overall performance across key metrics. Configuration w05 when applied to the test set performed similarly to its 100-epoch run with some minor variation. The final w05 model had a median silhouette score of 0.3365 (+0.0226 from w05 100 epoch run; see Table 4), stability ARI of 0.6657 (-0.2016), FCT cross-sample P@10 of 0.1521 (-0.0688), Location-probe macro-F1 of 0.3339 (-0.1277), source-image-enrichment@10 of 3.4261 (+0.8239) and an effective rank of 28.6195 (-6.8749).



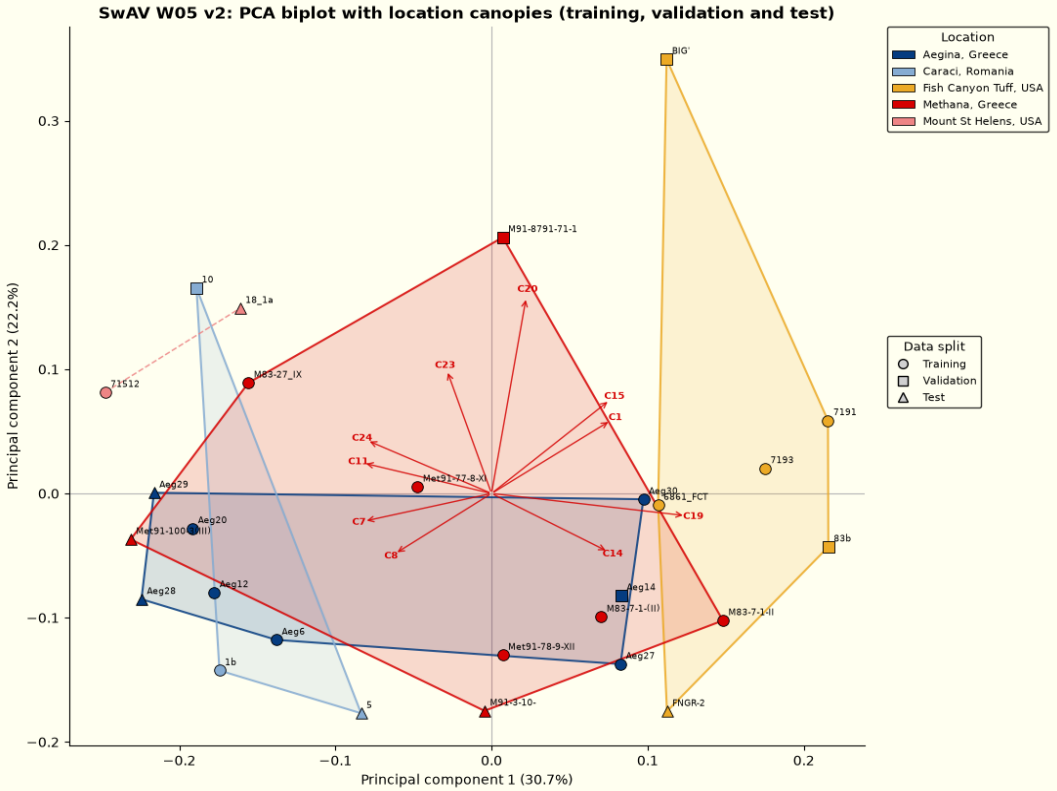


Figure 6 *– Cluster count and proportion by sample (Top). PCA biplot of all samples with canopies covering the area created by samples from each location (Bottom). Both samples created after the test run and therefore include the test samples.*

4. Discussion

Comparison of baselines shows differing self-supervised objectives create varying representations regardless of the shared use of the ResNet-18 backbone, preprocessing and evaluation procedures. SwAV produced the most defined clusters as evidenced by the highest silhouette score achieved by any of the baselines (0.2939), effective rank (48.75) and FCT cross-sample precision@10 (0.2362). This shows the models embeddings are combining comparatively distinct clusters and are using a comparatively large amount of the representation space in addition to having the strongest ability to retrieve related samples (FCT samples in this case) across samples. SimSiam had the strongest location-probe macro-F1 but also had a very low effective rank (7.89) suggesting high compression of the representation. Location probe is more diagnostic as it can be misled by non-texture image qualities like brightness. DeepCluster stood out for achieving the highest ARI stability index (0.6934) meaning DeepCluster clusters were reproducible across the most k-means initialisation. DeepCluster also had the highest (lower scores are better) source-image enrichment@10 (4.1829) meaning nearest neighbours were from the same cold-CL scan more often than any other model and more than can be ascribed to chance which suggests an issue like scan acquisition conditions, artefacts from segmentation or other scan characteristics may have partly caused its stable partitions. This is an example of a situation where stability metric cannot be taken alone as a model may perform consistently by separating images using image qualities completely unrelated to the geological texture it is intended to learn. Finally, the baseline SimCLR produced less competitive results than the other three models. However, it was notable for having a lower source-image enrichment than any other models and therefore can be seen as a counterpoint to DeepCluster and its poor source-image enrichment.

Not selecting a model purely based on loss was highlighted during the 30-epoch grid search configuration runs. Configurations such as s10 (SimSiam) had extremely promising negative cosine-similarity loss/objective primarily as an effect of the increase in learning rate. But the later 100 epoch run of the model chosen due to its high scores did not improve much further from its baseline model. In the SwAV grid search the best results were achieved when temperature increased from 0.05 to 0.1. This increase created wider and more balanced prototype use. The most stable and well-performing configurations from the prototypes = 250 and prototypes = 500 sections of the temperature = 0.1 half of the SwAV grid search were therefore chosen to be run for 100 epochs afterwards (w05 and w12). After 100 epochs w05 (learning rate = 0.3, temperature = 0.1, prototypes = 250) produced the strongest overall model seen during this study (median silhouette = 0.3139, ARI = 0.8672). Whilst w12 retained representational variance better than w05 (w12 effective rank = 42.91, w05 effective rank = 35.49) and had a stronger FCT-cross-sample precision@10. From this, w05 was selected as the final model having a strong ability to create reproducible, clear clustering and produced strong overall results on the unseen test data, slightly improving silhouette score compared to the 100-epoch optimised run.

Inspecting the clusters produced (Figure 5) suggests crystals are grouped at least partially by texture. Crystals with similar zoning, shapes and internal heterogeneities seem to be grouped together often. Therefore, it seems likely the SSL models are learning texturally and therefore volcanologically relevant information. However, clusters cannot yet be assigned with confidence to specific magmatic processes, and no expert-labelled benchmark study exists for comparison. Additionally, it appears some of the learnt groupings may have relied on image brightness, and therefore cold-CL intensity, which could reflect

compositional differences between the crystals, but differences in image generation conditions cannot be ruled out although efforts were made to regulate the production of the source scans. Certain groupings may result from other image acquisition traits. It is possible unrelated geological processes could produce cold-CL scans with similar appearances. Furthermore, the quality and accuracy of segmentation directly affect which information can be learnt by an SSL model. Crystals omitted by the segmentation model, single crystals being separated into multiple segmentation masks (Figure 1) and edge-touching objects all contribute to a bias towards crystals more easily identifiable by segmentation models. These issues may also affect the morphology and apparent abundance of clusters. Whilst square padding and black background standardise inputs there is a chance models could still learn unintended features from them such as background proportion. The augmentation during training may help to reduce these issues but it cannot be certified completely that only internal crystal texture is being used for learning. Cross-sample retrieval and source-image enrichment metrics are important then for judging if learned relationships go beyond individual scans and physical samples

Keeping crystals from each sample within the same split is useful for reducing leakage and producing a more rigorous assessment than splitting randomly at the single image level. Additionally, the unified dataset remains heavily imbalanced due to the influence of Methana's large proportion (55.2% total images and 58.7% training split). Contrasting this, Mt St Helens has just two samples with one used for training (71512) and another for the final test (18_1a) meaning before testing all analysis of and conclusions made about Mt St Helens are based solely on one sample. Measures like using Macro-F1 to give each location an equal weight do lower the influence of large, dominant locations like

Methana, but it is impossible for it to completely make up for a lack of diversity in raw scan data.

Only one ResNet-18 training seed was used for all models. Furthermore, performance would improve if models could be run for longer, as the number of epochs trained clearly will have a direct result. Testing other hyperparameters which were unable to be tested due to time constraints such as Sinkhorn epsilon, Sinkhorn iterations, projection dimension for SwAV and SGD momentum, batch size and projector/predictor dimensions for SimSiam. Additionally testing variations of already tested hyperparameters would likely improve performance for example temperature over 0.1 is projected to improve the occupied prototype fraction objective/loss for SwAV. The ResNet-18 backbone may not be the best possible shared architecture available. Finally, the lack of a dataset of similar plagioclase with labelled textures to test models on and assess whether a model has learnt meaningful textural information from the dataset would help assess model performance better.

Therefore, the main contribution made by this study is a method connecting previous segmentation studies (Nathwani and Peng, 2026), sample-aware dataset production, comparative metrics between SSL methods and SSL training and optimisation which culminate to produce promising proof-of-concept results. These results show SSL methods can order large datasets of crystals into recurring groups with shared visual information. However, it cannot yet be confirmed to have learnt solely textural information. Nor can it be claimed such methods yet perform at the level of a competent human observer let alone an expert observer. Despite these current limitations, these results provide a promising foundation for scalable, automated analysis of volcanic crystal textures using SSL.

5. Conclusions

PlagNetPlus combines sample-aware dataset production, SSL methods for learning unlabelled plagioclase crystal texture data and model optimisation techniques. SwAV consistently produced the best results with w05 becoming the final model and being tested on unseen data producing the highest silhouette score (0.3365) and therefore the clearest clusters seen during the study. Although, some of its other metrics declined somewhat compared to the optimised 100-epoch run. Visually inspecting the produced clusters often showed strong examples of crystals of similar texture and morphology being grouped. This suggests SSL methods such as those applied here can learn visual and textural relationships without labelling. However, no benchmarks exist (human or model) to compare this performance to and multiple issues may affect the quality of these learned representations like, brightness, segmentation quality (Figure 1) and conditions of image acquisition. Work in the future should aim then to reduce the effects of these limiting issues, incorporate more diverse, geographically balanced data. Broader and more thorough model optimisation such as grid searches with more hyperparameters, searching more extensive hyperparameter value ranges and higher epoch runs will also help improve performance.

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How generative AI was used: was used for understanding unknown terms and for summarising large sections of information during research.

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